EPIC-KITCHENS-100 Dataset Info

EPIC-KITCHENS-100

EPIC-KITCHENS-100 is a large-scale egocentric video dataset capturing daily activities in kitchen environments. It is multi-faceted, audio-visual, non-scripted recordings in native environments - i.e. the wearers' homes, capturing all daily activities in the kitchen over multiple days. Annotations are collected using a novel 'Pause-and-Talk' narration interface.



Characteristics

- 45 kitchens - 4 cities
- Head-mounted camera
- 100 hours of recording - Full HD
- 20M frames
- Multi-language narrations
- 90K action segments
- 20K unique narrations
- 97 verb classes, 300 noun classes
- 5 challenges

Short Term Anticipation (STA) task details

Task

Predict the next action (verb, noun) that will occur in the video, given a short observation of

the video segment.

Training input (strong supervision)

A set of trimmed action segments, each annotated with a (verb, noun) label. The model is provided with a short observation of the video segment (e.g., 1 second before the action starts).

Testing input

A set of trimmed unlabelled action segments, with a short observation of the video segment.

Splits

Develop models using team-defined train/validation splits of the provided training set; evaluate on the provided hidden test set.

Evaluation metrics

Class-mean Top-5 recall for the predicted next action (verb+noun), calculated for all segments as well as unseen participants and tail classes.

Action Recognition task details

Task

Assign a (verb, noun) label to a trimmed segment.

Training input (strong supervision)

A set of trimmed action segments, each annotated with a (verb, noun) label.

Testing input

A set of trimmed unlabelled action segments.

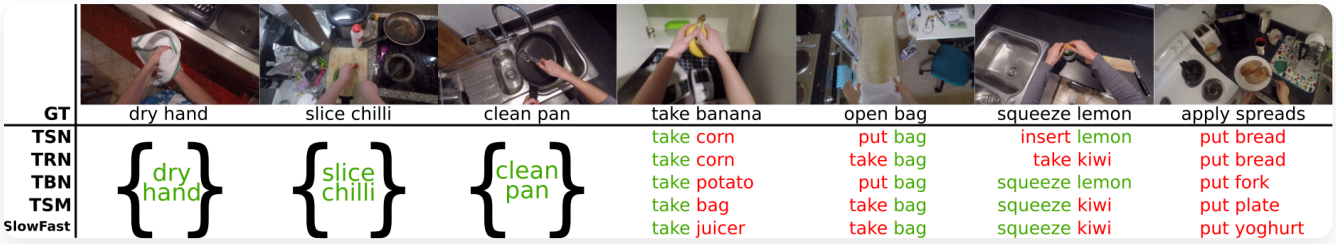
Splits

Develop models using team-defined train/validation splits of the provided training set; evaluate on the provided hidden test set.

Evaluation metrics

Top-1/5 accuracy for verb, noun and action (verb+noun), calculated for all segments as well

as unseen participants and tail classes.



Dataset Details

The EPIC-KITCHENS-100 dataset is an extension of the EPIC-KITCHENS-55 dataset. Videos are distinguished as follows:

- `PXX_YY.MP4` videos originate from EPIC-KITCHENS-55.
- `PXX_1YY.MP4` videos originate from the extension collected for EPIC-KITCHENS-100 (thus represent new videos).

Ground truth is provided for action segments as action/verb/noun labels along with the start and end times of the segment like so:

```
narration_id,participant_id,video_id,narration_timestamp,start_time
P01_01_0,P01,P01_01,00:00:01.089,00:00:00.14,00:00:03.37,8,202,open
P01_01_1,P01,P01_01,00:00:02.629,00:00:04.37,00:00:06.17,262,370,tu
P01_01_10,P01,P01_01,00:00:23.340,00:00:24.97,00:00:26.20,1498,1572
P01_01_100,P01,P01_01,00:07:57.919,00:07:59.75,00:08:00.88,28785,28
...
```

File Structure

Videos

The directory structure of the dataset is as follows.

```
Videos/
    ├── ..
    ├── videos
    |   └── P01
    |   |   ├── P01_01.mp4
    |   |   ├── P01_02.mp4
    |   |   ├── ...
    |   |   ├── ...
    |   |   └── P01_19.mp4
    |   └── P02
```

```
    │   │   ├── P02_01.mp4
    │   │   ├── P02_02.mp4
    │   │   ├── ...
    │   │   ├── ...
    │   │   └── P02_11.mp4
    │   └── ...
    │   └── P31
    └── annotations
        ├── EPIC_100_train.csv
        └── EPIC_100_test.csv
```

`EPIC_100_train.csv`

This CSV file contains the action annotations for the training set and contains 15 columns:

Column Name	Type	Example	Description
`narration_id`	string	`P01_01_0`	Unique ID for th
`participant_id`	int	`P01`	ID of the partici
`video_id`	string	`P01_01`	ID of the video
`narration_timestamp`	string	`00:00:01.089`	Timestamp of w
`start_timestamp`	string	`00:00:00.14`	Start time in `HH`
`stop_timestamp`	string	`00:00:03.37`	End time in `HH:`
`start_frame`	int	`8`	Start frame of t
`stop_frame`	int	`202`	End frame of th
`narration`	string	`open door`	Transcribed des
`verb`	string	`open`	Parsed verb fro
`verb_class`	int	`3`	Numeric ID of t
`noun`	string	`door`	First parsed nou
`noun_class`	int	`3`	Numeric ID of t
`all_nouns`	list of string (1 or more)	`[door]`	List of all parse
`all_noun_classes`	list of int (1 or more)	`[3]`	Numeric ID of a

`EPIC_100_test.csv`

This CSV file contains the action annotations for the course hidden test set and contains 9 columns, including the STA observation cutoff:

Column Name	Type	Example	Description
`narration_id`	string	`P01_01_11`	Unique ID for the segment as a st
`participant_id`	int	`P01`	ID of the participant (unique per p
`video_id`	string	`P01_11`	ID of the video where the segmer
`narration_timestamp`	string	`00:00:00.560`	Timestamp of when the original r
`start_timestamp`	string	`00:00:00.00`	Start time in `HH:mm:ss.SS` of the
`stop_timestamp`	string	`00:00:01.89`	End time in `HH:mm:ss.SS` of the a
`start_frame`	int	`1`	Start frame of the action.
`stop_frame`	int	`113`	End frame of the action.

The public hidden-test event table additionally contains `sta_observation_end_seconds`, the action start time minus 1.0 seconds. Negative values mean no STA video history is available.

Citation

Part of the content is adapted from the github repository associated with the dataset and available here.

```
@ARTICLE{Damen2022RESCALING,
           title={Rescaling Egocentric Vision: Collection, Pipeline
           author={Damen, Dima and Doughty, Hazel and Farinella, Gi
           and Ma, Jian and Kazakos, Evangelos and Moltisanti, Davi
           and Perrett, Toby and Price, Will and Wray, Michael},
           journal   = {International Journal of Computer Vision (I
           year      = {2022},
           volume = {130},
           pages = {33–55},
           Url       = {https://doi.org/10.1007/s11263-021-01531-2}
}
```

```
@INPROCEEDINGS{Damen2018EPICKITCHENS,
   title={Scaling Egocentric Vision: The EPIC-KITCHENS Dataset},
   author={Damen, Dima and Doughty, Hazel and Farinella, Giovanni M
           Furnari, Antonino and Kazakos, Evangelos and Moltisanti,
           and Perrett, Toby and Price, Will and Wray, Michael},
   booktitle={European Conference on Computer Vision (ECCV)},
   year={2018}
```